DDCA PROJECT

SMART STREET LIGHT AUTOMATION

A digital logic-based system for energy-efficient street lighting

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WHY AUTOMATE STREET LIGHTING?

THE PROBLEM

Manual or fixed-time lighting wastes power during daylight or when streets are empty. Traditional lights stay ON when not needed.

OUR SOLUTION

Use an LDR-based night signal and an IR motion signal as digital inputs to a logic gate to decide when lights turn ON.

PROJECT OBJECTIVES

Build the Boolean model, create its circuit in Logisim, and verify input combinations. *(Scope: Simplified digital prototype)*.

INPUTS, OUTPUTS AND THE AND GATE

INPUT A: NIGHT

A = 1: Night-time

A = 0: Daytime

INPUT B: MOTION

B = 1: Movement

B = 0: No movement

OUTPUT Y: LIGHT

Y = 1: Light ON

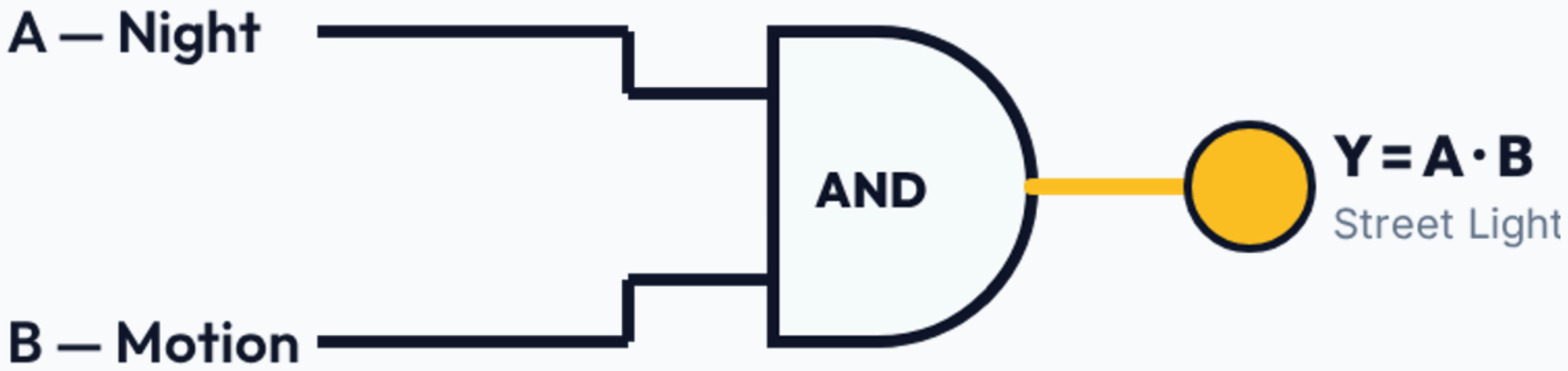
Y = 0: Light OFF

$$Y = A \cdot B$$

Why AND? The light should turn ON only when it is dark AND movement is detected.

LOGIC CIRCUIT DIAGRAM

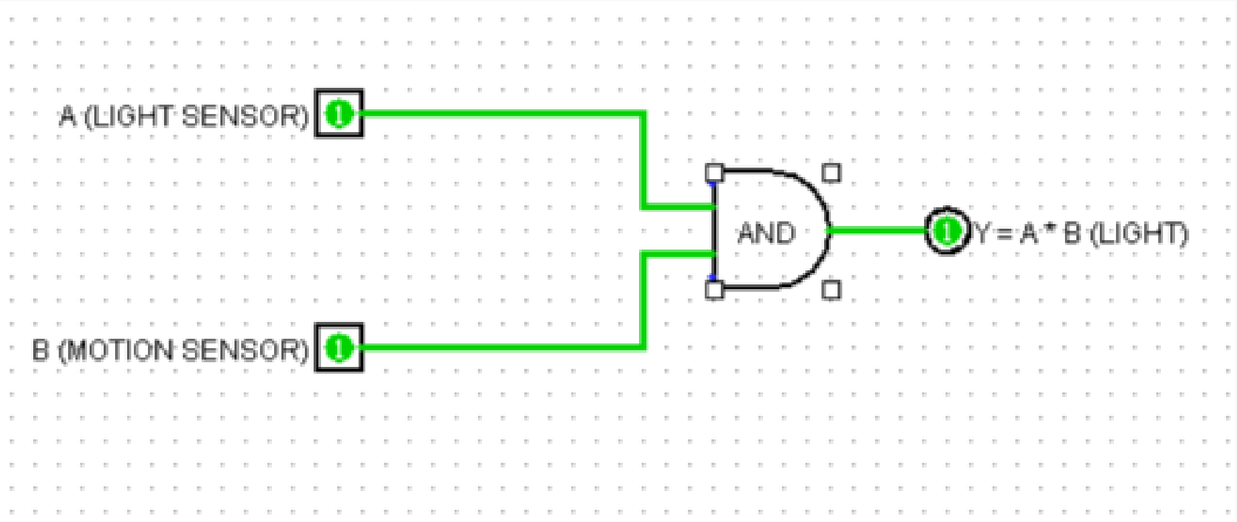
Digital model of the street-light decision



The output becomes **1** only for **A = 1** and **B = 1**.

IMPLEMENTING THE CIRCUIT

1. Open Logisim and create a new circuit.
2. Place two input pins and label them A (Night) and B (Motion).
3. Place one 2-input AND gate from the Gates library.
4. Connect A and B to the two AND-gate inputs.
5. Connect the gate output to an output pin or LED labelled Y (Street Light).



TRUTH TABLE AND EXPECTED OUTPUT

Night (A)	Motion (B)	Output Y	Street-light status
0	0	0	OFF — Daytime, no motion
0	1	0	OFF — Daytime, motion detected
1	0	0	OFF — Night, no motion
1	1	1	**ON — Night, motion detected**

Expected result: the output is HIGH in exactly one of the four cases—when both A and B are 1.

07 • CONCLUSION & APPLICATIONS

APPLICATIONS & IMPROVEMENTS

APPLICATIONS

- ✓ Campus pathways
- ✓ Residential streets
- ✓ Parking areas
- ✓ Low-traffic roads

ADVANTAGES

- → Reduces unnecessary lighting
- → Simple and low-cost logic
- → Easy to test in simulation

FUTURE IMPROVEMENTS

- Add a timer or delay
- Adjust sensitivity
- Use a microcontroller
- Add real sensor interfaces

CONCLUSION

A simple logic gate can make street lighting smarter.

Practical note: The prototype models decisions digitally (0/1). A real system requires sensor conditioning, a suitable lamp driver/relay, electrical protection, and outdoor testing.

THANK YOU

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